Benchmarking Principal Component and Random Projection dimensionality reduction methods for single cell RNA sequencing data

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Abstract

Principal Component Analysis (PCA) has long been a cornerstone in dimensionality reduction methods for high-dimensional data analyses, including genomics and single cell RNA sequencing (scRNA-seq) data. However, PCA algorithms performance typically deteriorate with data size, are sensitive to outliers and assume linearity. Recently, Random Projection (RP) methods have emerged as promising alternatives that address some of these issues. This study systematically evaluates multiple PCA algorithms, including Singular Value Decomposition (SVD) and randomized SVD, in comparison to Sparse and Gaussian Random Projections algorithms. We focus on computational efficiency and the effectiveness of downstream analyses. To that end, we use Hierarchical and Spherical K-Means clustering applied to several real labeled and unlabeled scRNA-seq datasets to benchmark performance and quality metrics for the methods of this study. Our results demonstrate that RP methods not only outperform PCA in computational speed but also rival PCA in capturing data variability and clustering quality. This work aims to guide researchers in selecting optimal dimensionality reduction techniques by balancing computational performance and downstream analysis needs and quality. By providing a thorough benchmarking for PCA and RP methods, we aim to we aim to facilitate a better understanding of their robustness, scalability, and interpretability of latent space dimensions.

Background

Single-cell RNA sequencing (scRNA-seq) methods often struggle with the challenge of high-dimensional data. Random Projections is a powerful dimensionality reduction technique that efficiently reduces the dataset size while preserving the variability between data points. This method is not only faster than more computationally intensive techniques like PCA but also maintains the structural integrity of scRNA-seq data. Consequently, RP accelerates the preprocessing step, enhancing both the speed and efficiency of subsequent analyses such as clustering, thereby improving the overall interpretability of the data [1].

Methods

We used two datasets for our experiments. One dataset is a targeted PBMC Sourced from a healthy donor, processed with 10x Genomics kits, this dataset includes about 10,497 cells with a focus on immune-related genes, achieving 87.5% mapping confidence. Subsamples from 1,000 to 10,000 cells were created to compare the clustering efficacy of PCA and Random Projections across different sizes, assessing the methods' scalability and robustness [2]. Our second dataset is a sorted PBMC dataset sourced from a healthy donor. This dataset was downloaded from the SC1 tool and has 2882 cells and 7174 genes. In our experiment, it was processed using its full size for baseline comparisons [3]. Dimensionality Reduction: We implemented PCA (both full and randomized SVD solvers) and Random Projections (Sparse and Gaussian), adjusting the number of components from 50 to 1000 in steps of 50. This allowed us to examine the influence of component granularity on the data structure.

Clustering Techniques

- Hierarchical Clustering with complete linkage: Effective for scRNA-seq data, this method uses cosine distances to create a dendrogram, reflecting cell relationships and facilitating data exploration at various resolutions based on the angular similarity between gene expression profiles.
- Spherical K-Means: Tailored for high-dimensional scRNA-seq data, Spherical K-Means clusters cells based on cosine distances, emphasizing directional similarity over magnitude. This method is especially useful for grouping cells with similar transcriptional directions in sparse datasets

Assessment Metric:

- Dunn Index (DI): This metric assesses the quality of clustering by the ratio of the smallest distance between observations in different clusters to the largest intra-cluster distance. It is computed as: $DI = \frac{\min_{(1 \le i \le j \le n)} d(i,j)}{\max_{(1 \le k \le n)} d'(k)}$

where i, j, and k are each indices for clusters, d measures the inter-cluster distance and d' measures the intra-cluster difference

Results – Locality Preservation

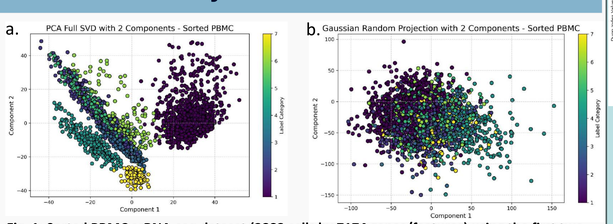

Fig. 1: Sorted PBMC scRNA-seq dataset (2882 cells by 7174 genes/features) using the first two components of each method. a Using PCA with full SVD . **b** Using Gaussian Random Projection. Some overlap is observed in RP more than in PCA using only the first two components.

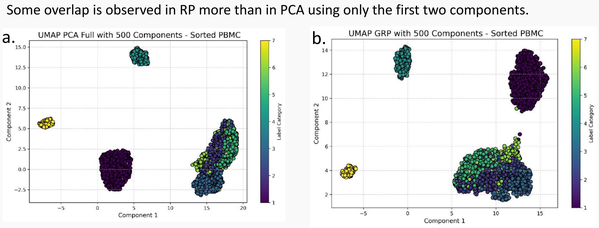

Fig. 2: Projecting the sorted PBMC dataset with reduction of 7174 dimensions to 500 dimensions into UMAP two-dimensional space. This figure shows the locality preservation characteristics of Random Projection followed by UMAP which compares to PCA. Panel a) shows PCA with full SVD method used to project the data in UMAP space. **b** Using Gaussian Random Projection.

Results – Time Comparisons

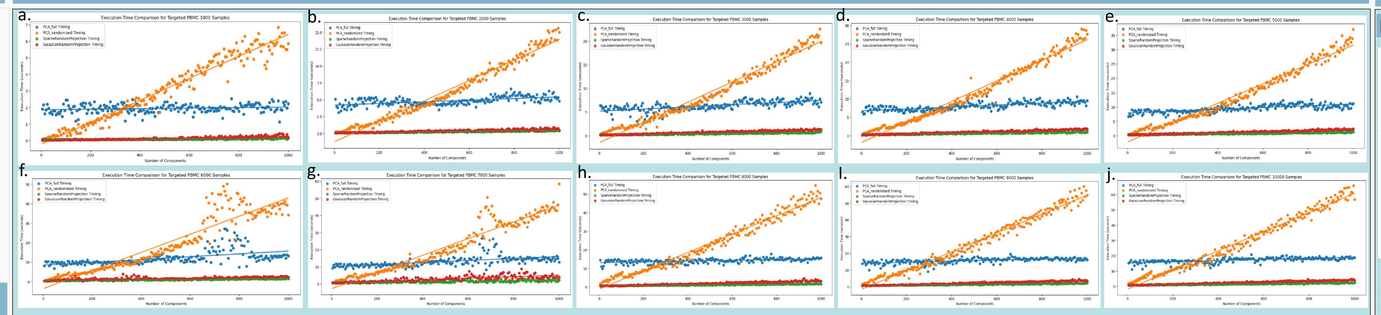

Fig. 3: Each plot corresponds to a certain dataset size for the Targeted PBMC dataset. The execution time vs the number of components used for each of the dimensionality reduction techniques used in this experiment. a Execution time for 1000 samples dataset **b** Execution time for 2000 samples dataset ..... **j** Execution time for 10000 samples dataset

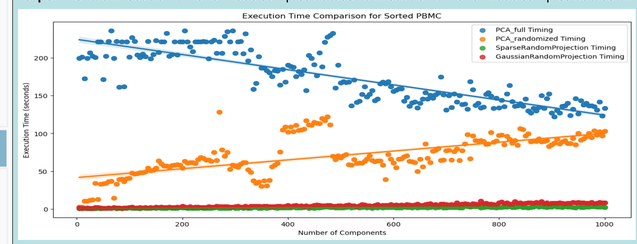

Fig. 4: The execution time vs the number of components for each of the dimensionality reduction techniques for the sorted PBMC dataset.

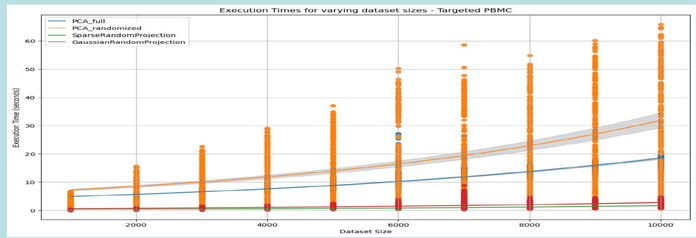

Fig. 5: The execution time vs the varying dataset sizes ranging from 1000 samples to 10000 samples with steps of 1000 for each of the dimensionality reduction techniques for the Targeted PBMC. Each dot represents a number of components, we varied the number of components from 5 to 1000 with steps of 5.

Full PCA uses the complete singular value decomposition (SVD), making its computational complexity $O(\min(n^2 \times d, n \times d^2))$, where n is the number of observations and d is the number of dimensions.

Randomized PCA offers a more computationally efficient alternative by approximating the decomposition, which reduces the complexity to $O(n \times d \times k + n \times k^2 + k^3)$ with k being the number of principal components retained.

Sparse Random Projection (SRP) and Gaussian Random Projection (GRP) both reduce dimensionality by projecting data onto a random lower-dimensional subspace. The time complexity for both is $O(n \times d \times k)$.

SRP utilizes a sparse random matrix, which typically leads to faster computations due to the lower number of non-zero entries.

GRP uses a dense matrix drawn from a Gaussian distribution and is more robust in preserving the distances but at the cost of higher computational overhead compared to SRP.

Results – Dunn Index - SphericalKMeans

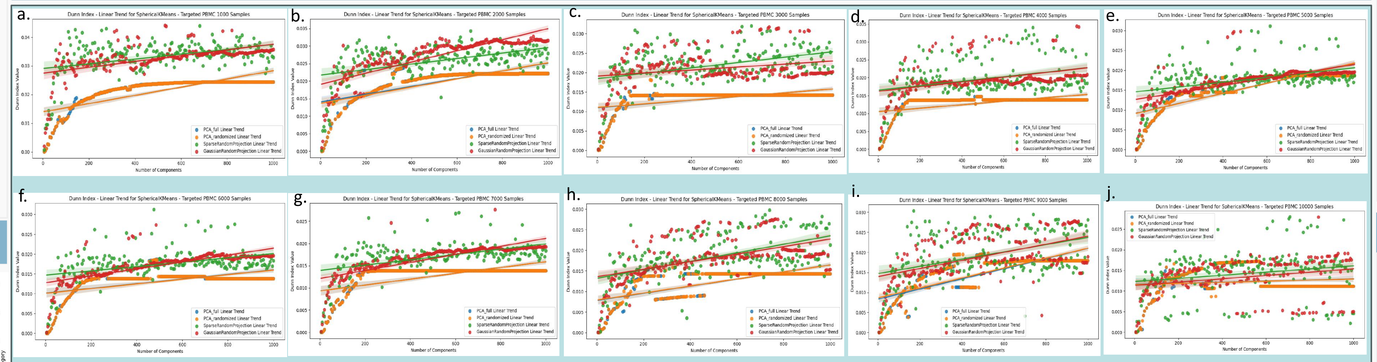

Fig. 6: Each plot corresponds to a certain dataset size for the Targeted PBMC dataset. Dunn Index vs the number of components used for each of the dimensionality reduction techniques used in this experiment. A higher Dunn Index value is indicative of a better clustering. a Dunn Index for 1000 samples dataset **b** Dunn Index for 2000 samples dataset ..... **j** Dunn Index for 10000 samples dataset

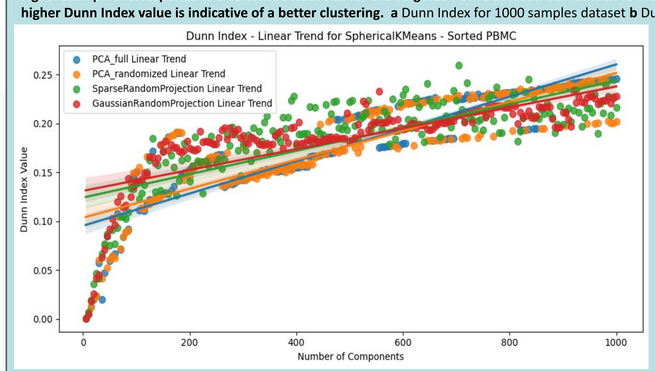

Fig. 7: Dunn Index vs the number of components for each of the dimensionality reduction techniques for the sorted PBMC dataset.

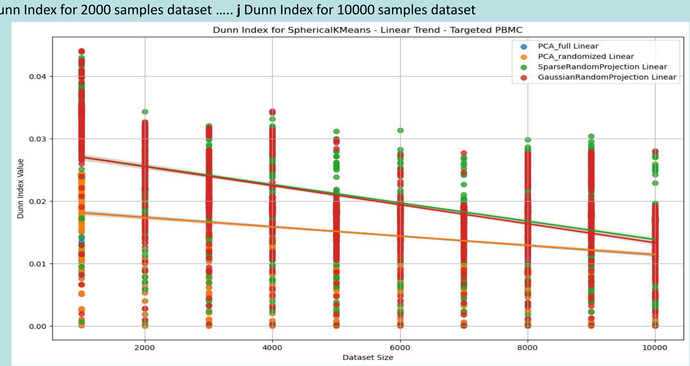

Fig. 8: Dunn Index vs the varying dataset sizes ranging from 1000 samples to 10000 samples with steps of 1000 for each of the dimensionality reduction techniques for the Targeted PBMC. Each dot represents number of components, we varied the number of components from 5 to 1000 with steps of 5.

Results – Dunn Index Continued

Hierarchical Clustering

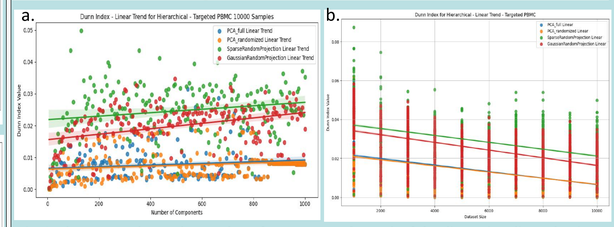

Fig. 9: The Dunn Index vs the number of components for each of the dimensionality reduction techniques for the Targeted PBMC dataset with Hierarchical Clustering . A higher Dunn index is a better clustering. a 10000 samples dataset **b** Varying dataset sizes 1000-10000 samples

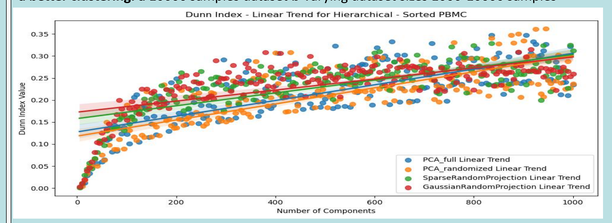

Fig. 10: The Dunn Index vs the number of components for each of the dimensionality reduction techniques for the Sorted PBMC dataset with Hierarchical Clustering. A higher Dunn index is a better clustering.

Discussion

Our analysis indicates that Random Projections (RP) consistently outperforms Principal Component Analysis (PCA) in terms of computational efficiency. We evaluated two variants of RP—Sparse Random Projections and Gaussian Random Projections—against two forms of PCA: PCA employing full Singular Value Decomposition (SVD) and PCA using randomized SVD. Our findings reveal that across both the Targeted PBMC dataset and the Sorted PBMC dataset, RP demonstrates a significant advantage in speed over PCA.

Figure 3 illustrates that PCA with randomized SVD outperforms PCA with full SVD up to a specific number of components; beyond this threshold, the performance of the randomized version deteriorates, taking considerably more time than its full SVD counterpart. Conversely, Figure 4 highlights the impact of dataset complexity on computation time, showing substantial increases for all algorithms when processing the Sorted PBMC dataset, which includes 7174 genes/features. In this context, PCA's performance lag is particularly pronounced.

Figure 2b showcases RP's ability to preserve locality when projecting the Sorted PBMC dataset into a UMAP two-dimensional space in a manner comparable to PCA. This aspect is crucial as locality preservation is essential for maintaining the integrity of data relationships in reduced dimensions, which is particularly beneficial for downstream analysis such as clustering and visualization.

Moreover, our evaluation of clustering effectiveness using the Dunn Index further supports the usage of RP. Both Gaussian and Sparse RP methods showed more effective clustering than both PCA variants, as shown across multiple clustering algorithms including Spherical K-Means (Figures 6, 7, and 8) and Hierarchical Clustering (Figures 9 and 10).

Conclusion

In conclusion, our study underscores the efficiency and effectiveness of Random Projections over Principal Component Analysis in handling high-dimensional data, particularly in terms of computational speed, clustering quality, and the preservation of data locality. RP not only offers a faster alternative but also maintains competitive performance in clustering outcomes and effectively preserves the geometric relationships of the dataset when reduced to lower dimensions, comparable to PCA. These findings suggest that RP could be a more viable option for data reduction in large-scale genomic studies, where speed, accuracy, and the integrity of data relationships are critical. Future research should explore the scalability of RP in other types of large datasets and consider the trade-offs between the number of components used and the quality of the data representation.

References

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